

Kowshika S

Data Analyst | Data Scientist

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PROFILE

Data Science graduate with hands-on experience in **Python, SQL, Machine Learning, Deep Learning, and Data Analysis**, achieving 99.82% accuracy in traffic sign classification. Built three end-to-end projects involving data preprocessing, feature engineering, computer vision, and dashboard development using Streamlit. Seeking opportunities as a Data Scientist, Data Analyst, or Machine Learning Engineer.

SKILLS

Python | SQL | Pandas | NumPy | Data Cleaning | Data Preprocessing | EDA | Feature Engineering | Excel | Power BI | Matplotlib | Seaborn | Streamlit | Scikit-learn | Classification | Regression | Predictive Modeling | TensorFlow | Keras | CNN | MobileNetV2 | OpenCV | Jupyter Notebook | VS Code | Git | GitHub

PROJECTS

AI-Powered Smart Mobility System

Key Skills: Python, TensorFlow, Keras, Scikit-learn, OpenCV, Streamlit

- Created an AI-powered Smart Mobility System using Machine Learning and Deep Learning to enhance road safety and transportation efficiency.
- Designed a CNN-based Traffic Sign Classification model using **50,000+ labeled images across 43 classes**, achieving **99.82% accuracy**.
- Implemented a Road Risk Prediction model with **87.50% accuracy** and integrated Sentiment Analysis to classify road-related complaints.
- Combined traffic sign recognition, road risk prediction, and sentiment analysis into a unified AI solution supporting real-time predictions.

Driver Drowsiness Detection

Key Skills: Python, TensorFlow, Keras, MobileNetV2, OpenCV

- Engineered a real-time Driver Drowsiness Detection system using Computer Vision and Deep Learning.
- Trained a MobileNetV2 model on **2,000 labeled images across four classes**.
- Achieved **98.97% accuracy, 99.01% precision, 98.97% recall, and 98.97% F1-score**.
- Enabled real-time webcam-based drowsiness detection and driver alert generation using OpenCV.

Job Acceptance Prediction System

Key Skills: Python, Pandas, Scikit-learn, MySQL, Streamlit

- Created a Machine Learning solution to predict candidate job-offer acceptance using **51,500 records with 26 features**.
- Performed **data cleaning, preprocessing, feature engineering, and EDA** to prepare the dataset for model training.
- Evaluated **Logistic Regression, Decision Tree, and Random Forest** algorithms, with Random Forest achieving **88.63% accuracy**.
- Deployed a **Streamlit dashboard** for real-time prediction and stored candidate data in **MySQL**.

EDUCATION

Bachelor of Engineering - Computer Science and Engineering, Anna University

Knowledge Institute of Technology, Salem | 2016 – 2020 | 68.2%

CERTIFICATIONS

- IIT-M Advanced Programming Professional & Master Data Science, *IIT-M GUVI HCL* | Feb 2026 – May 2026
- Intro to Python for Data Science, DataCamp, 2019
- Certificate of Merit in Data Structures, Texas Instruments India, 2018